# Hita Kambhamettu

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## Research Interests

I study human-AI collaboration for frontier knowledge extension: how AI can help people develop new knowledge that builds on what already exists (i.e. in scientific discovery). My work combines (1) building structured representations of existing knowledge that people and models can explore and reason over, with (2) developing and evaluating human-AI systems for understanding and producing new knowledge.

## Honors

2026 **Best Paper Honorable Mention at UIST**,
for *Attribution Gradients: Incrementally Unfolding Citations for Critical Examination of Attributed AI Answers*

2024 **National Science Foundation**, Graduate Research Fellowship

2023 **National Science Foundation**, Graduate Research Fellowship Honorable Mention

2021 **National Science Foundation**, Research Experience for Undergraduates Scholarship

## Education

Sept 2022–Present **University of Pennsylvania** – Philadelphia, PA
*PhD* in Computer and Information Science
*Advisor*: Andrew Head , Lyle Ungar
*Research Areas*: Human-computer interaction, Natural language processing

May 2022 **Carnegie Mellon University** – Pittsburgh, PA
*BS* in Information Systems
*Minors* in Computer Science and Machine Learning
*Advisor*: Rohan Padhye

## Research Experience

Summer 2026 — **Microsoft Research Cambridge** – Cambridge, UK
*Research Focus:* Studying how knowledge embedded in AI agents' answers shapes what people incidentally learn from them.
*Advisor*: Advait Sarkar (People-Centric AI team)

Summer 2025 — **Allen Institute for AI** – Seattle, WA
*Research Focus:* Augmenting scientific ideation with an intelligent tool to support literature-based changes to research ideas.
*Advisors*: Pao Siangliulue, Joseph Chee Chang, Aakanksha Naik, Jonathan Bragg

Fall 2022–Present — **Penn HCI Lab** – Philadelphia, PA
*Research Focus:* Building intelligent reading and writing interfaces that align AI systems with human information reasoning processes.
*Advisors*: Andrew Head, Lyle Ungar

Summer 2021 — **PASTA Lab** – Pittsburgh, PA
*Research Intern (REU)*
*Research Focus:* Leveraging deep learning techniques to investigate the behavior of fuzzer-generated code.
*Advisors*: Rohan Padhye, Vincent Hellendoorn

Summer 2020 — **Human-Centered Robotics Lab** – Seattle, WA
*Research Intern*
*Research Focus:* Investigating how programming systems can support how end users naturally express robot programs.
*Hosts*: Vinitha Ranganeni, Maya Cakmak

## Publications, Articles, and Preprints

** denotes equal contribution*

under review — **Guiding AI Agents to Generate Conjectures by Encoding How Theorems Build on One Another**
**Hita Kambhamettu**, Daniel Krashen, Lyle Ungar.
*under review*.

under review

**More Than What You Asked For: AI Answers Enriched with Task-Adjacent Knowledge Improve Users' Evaluation of What Those Answers Support**
**Hita Kambhamettu**, Philippe Laban, Christian Poelitz, Advait Sarkar.
*under review*.

NeurIPS Math-AI workshop

**Theorem Evolution Graphs: A Structured Representation of How Theorems Build on One Another**
**Hita Kambhamettu**, Daniel Krashen, Lyle Ungar.
*NeurIPS Math-AI workshop*.

UIST 2026

**Attribution Gradients: Incrementally Unfolding Citations for Critical Examination of Attributed AI Answers**
**Hita Kambhamettu**, Alyssa Hwang, Philippe Laban, Andrew Head.
*Best Paper Honorable Mention.*
*ACM Symposium on User Interface Software and Technology* .

UIST 2026

**LitPivot: Developing Well-Situated Research Ideas Through Dynamic Contextualization and Critique within the Literature Landscape**
**Hita Kambhamettu**, Bhavana Dalvi Mishra, Andrew Head, Jonathan Bragg, Aakanksha Naik, Joseph Chee Chang, Pao Siangliulue.
*ACM Symposium on User Interface Software and Technology*.

UIST 2026

**Explorable Proofs: Making Written Proofs Explorable by Grounding Them in Formal Representations**
**Hita Kambhamettu**, Will Crichton, Sean Welleck, Harrison Goldstein, Andrew Head.
*ACM Symposium on User Interface Software and Technology*.

preprint

**Connecting the Dots: Surfacing Structure in Documents through AI-Generated Cross-Modal Links**
Alyssa Hwang, **Hita Kambhamettu**, Yue Yang, Ajay Patel, Joseph Chee Chang, Andrew Head.
*arXiv preprint*.

FAccT 2026

**Everyday Auditing of TikTok's Generative AI Manga Filter**
Ro Encarnación, Luis Morales-Navarro, **Hita Kambhamettu**, Danaé Metaxa.
*ACM Conference on Fairness, Accountability, and Transparency*.

CHI EA 2026

**Towards Caregiver-Centric Health Technology in Intricate Care Situations: A Case Study of Childhood Asthma Care in the Local Community**
Jamie Flores, Anthonia Odarkor Odartei, Levia Airall Sutton, **Hita Kambhamettu**, Sonya Sanders, Kenrick Cato, Aviv Landau, Andrew Head.
*ACM Conference on Human Factors in Computing Systems (Extended Abstracts)*.

SSWR 2026 **Integrating Community Knowledge into Electronic Health Records: Reimagining Childhood Asthma Care**.
Aviv Landau, **Hita Kambhamettu**, Anthonia Odartei, Levia Sutton, Sonya Sanders, Andrew Head, Kenrick Cato.
*Conference of the Society for Social Work and Research (Oral Presentation)*

CSCW 2025 **PAL: Designing Conversational Agents as Scalable, Cooperative Patient Simulators for Palliative-Care Training**
Neil Sehgal*, **Hita Kambhamettu***, Allen Chang*, Andrew Zhu*, Lyle Ungar, Sharath Chandra Guntuku.
*Computer-Supported Cooperative Work and Social Computing (posters track)*.

CHI EA 2025 **Exploring Socio-Cultural challenges and opportunities in designing mental health chatbots for adolescents in India**
Neil Sehgal, **Hita Kambhamettu**, Sai Preethi Matam, Lyle Ungar, Sharath Chandra Guntuku.
*ACM Conference on Human Factors in Computing Systems*.

CHI EA 2025 **Traceable Texts and Their Effects: A Study of Summary-Source Links in AI-Generated Summaries.**
**Hita Kambhamettu**, Jamie Flores, Andrew Head.
*ACM Conference on Human Factors in Computing Systems*.

AMIA 2024 **Patient-centered Evaluation of AI Answers to Genetic Counseling Questions**
**Hita Kambhamettu***, Yidi Huang*, Kevin Johnson, Angela Bradbury.
*American Medical Informatics Association*.

CHI 2024 **Explainable Notes: Examining How to Unlock Meaning in Medical Notes with Interactivity and Artificial Intelligence.**
**Hita Kambhamettu**, Danaé Metaxa, Kevin Johnson, Andrew Head.
*ACM Conference on Human Factors in Computing Systems*.

IIS 2024 **Accelerating Scientific Paper Skimming with Augmented Intelligence Through Customizable Faceted Highlights.**
Raymond Fok, Luca Soldaini, Cassidy Trier, Erin Bransom, Kelsey MacMillan, Evie Cheng, **Hita Kambhamettu**, Jonathan Bragg, Kyle Lo, Marti A. Hearst, Andrew Head, Daniel S. Weld
*ACM Transactions on Interactive Intelligent Systems.*

AAAI 2024 **Knowledge-Grounded Medical Dialogue Generation for Genetic Counseling Regarding Alzheimer's Risk.**
**Hita Kambhamettu***, Yidi Huang*, Kevin Johnson, Angela Bradbury.
*Workshop on Health Intelligence (co-located with AAAI)*.

UIST 2023 **FFL: A Language and Live Runtime for Styling and Labeling Typeset Math Formulas.**
Zhiyuan Wu, Jiening Li, Kevin Ma, **Hita Kambhamettu**, Andrew Head.
*ACM Symposium on User Interface Software and Technology*.

IUI 2023 **Scim: Intelligent Skimming Support for Scientific Papers.**
Raymond Fok, **Hita Kambhamettu**, Luca Soldaini, Jonathan Bragg, Kyle Lo, Marti A. Hearst, Andrew Head, Daniel S. Weld.
*ACM Conference on Intelligent User Interfaces*.

MSR 2022 **On the Naturalness of Fuzzer-Generated Code.**
**Hita Kambhamettu***, John Billos*, Carolyn Oluw Oluwaseun-Apo*, Rohan Padhye, Vincent Hellendoorn
*Proceedings of the 19th International Conference on Mining Software Repositories*.

PLATEAU 2022 **Collecting Insights about How Novice Programmers Naturally Express Programs for Robots.**
**Hita Kambhamettu**, Michael Jae-Yoon Chung, Vinitha Ranganeni, Patricía Alves-Oliviera
*Workshop on the intersection of HCI and PL*.

CCNC 2022 **ImPos: An Image-Based Indoor Positioning System.**
Yunzhi Li, **Hita Kambhamettu**, Yidan Hu, Rui Zhang
*IEEE Annual Consumer Communications & Networking Conference*.

## Teaching

Fall 2023 **Head Teaching Assistant**, CIS 5120 Introduction to Human-Computer Interaction
Instructor: Danaé Metaxa (University of Pennsylvania)

Spring 2022 **Teaching Assistant**, 15-694 Cognitive Robotics
Instructor: David Touretzky (Carnegie Mellon University)

Spring 2022 **Teaching Assistant**, 67-364 Practical Data Science
Instructor: Raja Sooriamurthi (Carnegie Mellon University)

Fall 2021 **Teaching Assistant**, 17-313: Foundations of Software Engineering
Instructor: Rohan Padhye (Carnegie Mellon University)

## Invited Talks

November 2026 — **UIST Doctoral Symposium**
Structured Knowledge Representations for Epistemic Exploration in Human-AI Interactions

November 2025 — **Adobe Research**
Knowledge Scaffolds for Reasoning with AI

August 2025 — **Tableau Research**
AI-Augmented Reasoning: Knowledge Scaffolds for Complex Information Consumption

June 2025 — **Allen Institute for AI**
Supporting Pivots in Research Ideas through {Pro, Retro, Inter}-active Agents

Oct 2024 — **Penn CIS Health, Healthcare, and Technology**
Designing Interactive Agents for Answering Difficult Medical Questions

Feb 2024 — **Penn Medicine Patient Advisory Council**
Explainable Notes: How to Unlock Meaning in Medical Notes with Artificial Intelligence

Oct 2023 — **Penn ASSET/IBI Symposium on Trustworthy AI for Health Care**
Knowledge-Grounded Medical Dialogue Generation

June 2023 — **University of Pennsylvania GEMS Seminar**
Human-centered Opportunities for Building Effective Conversational Agents

## Selected Media and Impact

Oct 23, 2026 — **LitPivot: Developing research ideas through dynamic contextualization within the literature landscape**
Kyle Wiggers, *Ai2 Blog Post*.

Apr 12, 2024 — **Penn HCI & NLP Students Awarded NSF GRFP**
Alexandria Moraschi, *Penn CIS Newsletter*.

Jan 11, 2024 — **Coming Together To Transform Tech**
Janelle Weaver, *Penn Engineering Magazine*.

Aug 21, 2023 — **Bringing Scientific Papers to Life**
Marti Hearst, *ICDAR 2023 Keynote*.

## Academic Service & Mentoring

2026 **Penn HCI Seminar**
*Organizer*, PennHCI

2025 **HCI + NLP Reading Group**
*Organizer*, PennNLP

2024 **Machine Learning for Health**
*Subchair*, Workflow and Review Committee

2024 **CHI, UIST, DIS**
*Peer Reviewer*

2024 **CMU Mentorship Program**
Veronica Pimenova (CMU Undergraduate)

2023 **AEOP Mentorship Program**
Evelyn Parra de Jesus (Arcadia University Undergraduate)

2023 **Penn GEMS Engineering, Science, and Math Camp**
*Guest Lecturer*

## Leadership

Fall 2023–Present *PhD Rep.*, **Computer and Information Science Admitted Students Open House**

Fall 2022–Present *Organizer*, **Penn CIS PhD Mentorship Program**

Fall 2022–Present *PhD Rep.*, **Penn Engineering Graduate Student Government**

Fall 2023–Fall 2024 *PhD Rep.*, **Department of Computer and Information Science Office Committee**